

Create data set compatible with ShinyLib

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In a first step, the results given by the Nextflow pipeline are loaded.

```
load("../results/fitness/result.Rdata")
load("../input/CRISPR_2022_dataSet_forShinyLib/CRISPRi_library_2022.Rdata")

annotation <- unique(CRISPRi_library_2022[,c(1,26:33)])
DESeq_result_table <- DESeq_result_table %>% left_join(annotation)
DESeq_result_table$FoldChange <- 2^DESeq_result_table$log2FoldChange
CRISPRi_library_2025_Lys <- DESeq_result_table

CRISPRi_library_2025_Lys[grepl("nc_", CRISPRi_library_2025_Lys$sgRNA),]$sgRNA_type <- "non-coding"
CRISPRi_library_2025_Lys[grepl("nc_", CRISPRi_library_2025_Lys$sgRNA),]$protein <- "non-coding RNA"
CRISPRi_library_2025_Lys[grepl("nc_", CRISPRi_library_2025_Lys$sgRNA),]$process <- "non-coding RNA"
CRISPRi_library_2025_Lys[grepl("nc_", CRISPRi_library_2025_Lys$sgRNA),]$pathway <- "unknown"

df_ncRNAs <- as.data.frame(str_split_fixed(CRISPRi_library_2025_Lys[grepl("nc_", CRISPRi_library_2025_Lys$sgRNA),]$sgRNA_target, 3, 1))
df_ncRNAs$sgRNA_target <- CRISPRi_library_2025_Lys[grepl("nc_", CRISPRi_library_2025_Lys$sgRNA),]$sgRNA_target
df_ncRNAs <- unique(df_ncRNAs[,c(2,3)])
names(df_ncRNAs) <- c("locus", "sgRNA_target")
CRISPRi_library_2025_Lys <- rows_patch(CRISPRi_library_2025_Lys, df_ncRNAs, by="sgRNA_target")
ncRNA_annotation <- read_tsv("../input/ncRNAs_CRISPRi.csv")
CRISPRi_library_2025_Lys <- rows_patch(CRISPRi_library_2025_Lys, ncRNA_annotation, by="locus", unmatched="na")
names(df_ncRNAs) <- c("gene_name", "sgRNA_target")
df_ncRNAs <- tibble(df_ncRNAs)
CRISPRi_library_2025_Lys <- rows_patch(CRISPRi_library_2025_Lys, df_ncRNAs, by="sgRNA_target")

CRISPRi_library_2025_Lys[is.na(CRISPRi_library_2025_Lys$gene_name_short),]$gene_name_short <- CRISPRi_library_2025_Lys$sgRNA_target

list_with_locus_tags <- (grepl("sll", CRISPRi_library_2025_Lys$gene_name) | grepl("slr", CRISPRi_library_2025_Lys$gene_name))
CRISPRi_library_2025_Lys[!list_with_locus_tags,]$gene_name <- paste(CRISPRi_library_2025_Lys[!list_with_locus_tags,]$sgRNA_target, "non-coding")
ncRNAs <- subset(CRISPRi_library_2025_Lys, CRISPRi_library_2025_Lys$sgRNA_type=="non-coding")

save(CRISPRi_library_2025_Lys, file = "../R_results/CRISPRi_library_2025_Lys.Rdata")
```

Session info

```
## R version 4.4.2 (2024-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.5 LTS
```

```

##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.20.so; LAPACK version 3.10.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C
## [3] LC_TIME=sv_SE.UTF-8 LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=sv_SE.UTF-8 LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=sv_SE.UTF-8 LC_NAME=C
## [9] LC_ADDRESS=C LC_TELEPHONE=C
## [11] LC_MEASUREMENT=sv_SE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Stockholm
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] magrittr_2.0.3 lubridate_1.9.3 forcats_1.0.0 stringr_1.5.1
## [5] dplyr_1.1.4 purrr_1.0.2 readr_2.1.5 tidyr_1.3.1
## [9] tibble_3.2.1 ggplot2_3.5.1 tidyverse_2.0.0 knitr_1.49
##
## loaded via a namespace (and not attached):
## [1] bit_4.5.0 gtable_0.3.6 crayon_1.5.3 compiler_4.4.2
## [5] tidyselect_1.2.1 parallel_4.4.2 scales_1.3.0 yaml_2.3.10
## [9] fastmap_1.2.0 R6_2.5.1 generics_0.1.3 munsell_0.5.1
## [13] pillar_1.9.0 tzdb_0.4.0 rlang_1.1.4 utf8_1.2.4
## [17] stringi_1.8.4 xfun_0.49 bit64_4.5.2 timechange_0.3.0
## [21] cli_3.6.3 withr_3.0.2 digest_0.6.37 grid_4.4.2
## [25] vroom_1.6.5 rstudioapi_0.17.1 hms_1.1.3 lifecycle_1.0.4
## [29] vctrs_0.6.5 evaluate_1.0.1 glue_1.8.0 fansi_1.0.6
## [33] colorspace_2.1-1 rmarkdown_2.29 tools_4.4.2 pkgconfig_2.0.3
## [37] htmltools_0.5.8.1

```